

From ③, we know $\exists x (C(x) \wedge S(x))$ is True. By defn. of existential quantifier, there is a student a in the class s.t. $C(a) \wedge S(a)$ is True. From defn. of AND, $C(a)$ is True and $S(a)$ is True.

From ⑦, we know $\forall x (S(x) \Leftrightarrow \neg F(x) \wedge \neg J(x) \wedge \neg Se(x))$ is True.

By defn. of universal quantifier, for all students b in the class, $S(b) \Leftrightarrow \neg F(b) \wedge \neg J(b) \wedge \neg Se(b)$ is True. $\therefore S(a) \Leftrightarrow \neg F(a) \wedge \neg J(a) \wedge \neg Se(a)$ is True. By defn. of biconditional, $\neg F(a) \wedge \neg J(a) \wedge \neg Se(a)$ is True.

By defn. of AND, $\neg F(a)$, $\neg J(a)$ and $\neg Se(a)$ are all respectively True.

By defn. of negation, $\neg C(a)$ is False. By ~~defn~~ ⑪, we know $\forall x (M(x) \Leftrightarrow \neg C(x))$ is True. By defn. of universal quantifier, for all elements b in the domain, $M(b) \Leftrightarrow \neg C(b)$ is True. $\therefore M(a) \Leftrightarrow \neg C(a)$ is True. By defn. of biconditional, $M(a)$ is False. By defn. of negation, $\neg M(a)$ is True. By defn. of AND, $\neg M(a) \wedge \neg J(a)$ is True. By defn. of existential quantifier, $\exists x (\neg M(x) \wedge \neg J(x))$ is True.

d) Every student in the class is either a sophomore or a computer science major: $\forall x (S(x) \vee C(x))$

From ①, we know that $\exists x (M(x) \wedge F(x))$ is True. By defn. of existential quantifier, there exists at least one student a in the class s.t. $M(a) \wedge F(a)$ is True. By defn. of AND, $M(a)$ is True, $F(a)$ is True.

From ⑧, we know that $\forall x (F(x) \Leftrightarrow \neg S(x) \wedge \neg J(x) \wedge \neg Se(x))$ is True. By defn. of universal quantifier for all students b in the class, $F(b) \Leftrightarrow \neg S(b) \wedge \neg J(b) \wedge \neg Se(b)$ is True. $\therefore F(a) \Leftrightarrow \neg S(a) \wedge \neg J(a) \wedge \neg Se(a)$ is True. By defn. of biconditional, $\neg S(a) \wedge \neg J(a) \wedge \neg Se(a)$ is True. By defn. of AND, $\neg S(a)$, $\neg J(a)$, $\neg Se(a)$ are all True respectively. By defn. of negation, $\neg(\neg S(a))$ is False. By double negation law, $S(a)$ is False.